CIVL1113 Engineering
Mechanics and Materials

December 2024 Examination
Comments on student performance
Mechanics part

General: Mechanics part

- The performance of students varies greatly. In general, many students were able to solve the problem and showed basic understanding on the course contents in mechanics part.
- A problem of some students is the lack of understanding of internal forces in a structure, which is related to inability to construct bending moment diagram (BMD) and shear force diagram (SFD) diagram correctly. Looking at BMDs and SFDs for typical cases will help.
- Some students did not have the ability to spot problems from unusual results, e.g., convert between different unit systems, bending moment diagram should be zero at pin joints unless external moment applied, etc., and to follow up with further checking.
- Some students had difficulty in remembering the basic equations for the calculation, which had been emphasized many times in exams. Some basic equations are noted as very important equations and marked in red in lecture notes.
- Taking a look at last semester tutorial video recorded (content different from this year version) may help, especially for buckling.

Question 1

- The performance in this question was fine but varied.
- In Question 1(a), the performance was generally fine but there are still some students who did not work out the results clearly. This question of beam problem can be solved from previous course ENGG1300 knowledge.
- Question 1(b), students should be able to understand the force transfer in this structure consisting of 3 hinges. This is a three-hinge arch. Some students failed to calculate the reaction forces. Not all students follow the steps to obtain the results.
- Some students presented BMD and SFD of wrong shapes, suggesting inadequate understanding of the concepts.
- Some students showed inadequate understanding of the sign conventions in the construction of BMD and SFD as guided by the exam questions.

Question 2

- The overall performance in this question was fine. The performance varied.
- Some students had some difficulty in converting the unit systems in Q2(a).
- Some student used the shear force equation of rectangular beam, which is not correct.
- This Q2a has been carefully designed. If you use maximum normal stress and maximum shear stress into the von Mises equation, you may obtain a wrong conclusion. However, if you carefully calculate the von Mises stress at top flange, intersection between top flange and web and web, you will find the beam is safe using von Mises criterion.
- Some students used a wrong equation for Q2(b).
- For this Q2(b)(i), the stress transformation can be solved by either remembering the equation or using Mohr circle. However, there are some errors shown in both methods in some students.
- Some students did not well use the strength criteria to check the safety as requested by the exam paper.

Question 3

- The overall performance in this question was moderate. This problem is similar to a question in Homework 4. That question is more complicated than this exam problem. If student is able to solve Homework 4 independently, this exam problem should be straightforward.
- This buckling issue problem. Some students remembered Euler's equation incorrectly. Some students did not work out the effective length correctly for in-plane buckling and out-of-plane buckling correctly. Some students did not consider out-of-plane buckling, which was clearly stated in the exam paper that need to consider. For this example, when we say hinge, by default this is spherical hinge instead of cylinder hinge. In other words, the end are hinge for both in-plane and out-of-plane buckling. Some students was having difficulty in working out the unit convert.